

Math 166
Polar coordinates

Name: _____

Section Number: _____

Goal. Apply calculus to polar coordinates.

1. Plot the following polar coordinates: $(-7, \frac{\pi}{2})$, $(3, \frac{4\pi}{3})$, $(2, \pi)$, $(0, 2)$. Then write each polar coordinate in a different way (i.e. with a different value of r and/or θ). Lastly, from your graph or using the equations $x = r \cos(\theta)$ and $y = r \sin(\theta)$, convert the points from polar to rectangular coordinates.
2. Plot the following rectangular coordinates: $(-5, 0)$, $(1, -2)$, $(-1, 2)$, $(12, 5)$. Then, convert the points from rectangular to polar coordinates.

3. Convert $2x^2 + 5x^3 = 1 + xy$ into polar coordinates.

4. Convert $r = -8 \cos(\theta)$ into Cartesian/rectangular coordinates.

5. Plot $r = -8 \cos(\theta)$ for $0 \leq \theta \leq 2\pi$.

6. Graph the polar equation $r = 1 + 2 \cos(\theta)$.

7. Find all values of θ such that $0 \leq \theta \leq 2\pi$ and $r = 1 + 2 \cos(\theta) = 0$.

8. Determine the area of the inner loop of $r = 1 + 2 \cos(\theta)$.
Hint: Use Problems 6 and 7 to find the bounds on the integral.

9. Find the arc length of the polar equation $r = 2\theta$ for $0 \leq \theta \leq 1$.